1.

hamsini@HamsiniVanam: ~

```
 GNU nano 8.7.1
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

```
hamsini@HamsiniVanam:~$ nano prog100.c
hamsini@HamsiniVanam:~$ gcc prog100.c -o
gcc: error: missing filename after ‘-o’
hamsini@HamsiniVanam:~$ gcc prog100.c -o prog100
hamsini@HamsiniVanam:~$ ./prog100
Value of x = 10
Virtual address of x = 0x7ffe5bdd17e4
hamsini@HamsiniVanam:~$
```

2.

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

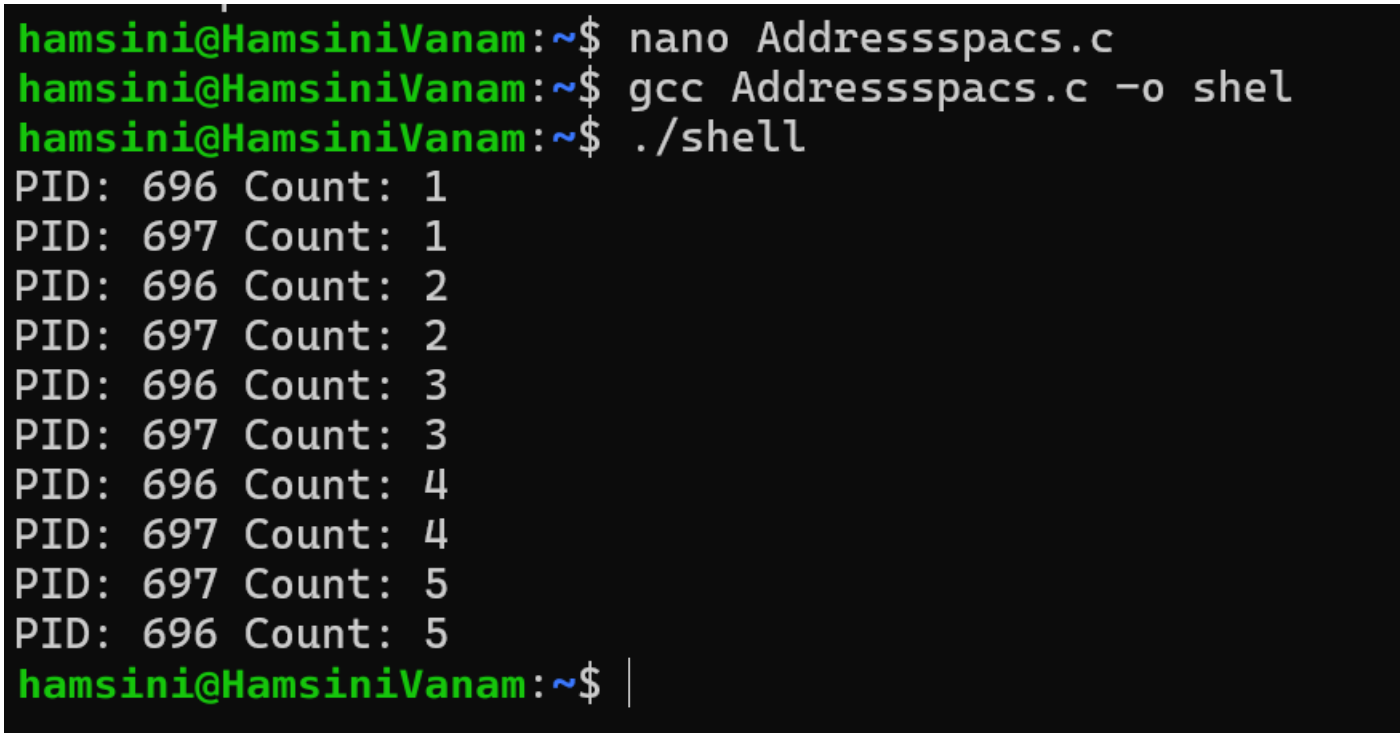
    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address   : %p\n", (void *)main);
    printf("Data address   : %p\n", (void *)&global);
    printf("BSS address    : %p\n", (void *)&bss_varia
    printf("Heap address   : %p\n", (void *)heap);
    printf("Stack address  : %p\n", (void *)&local);

    free(heap);

    return 0;
}
```


hamsini@HamsiniVanam:~$ nano Addressspacs.c
hamsini@HamsiniVanam:~$ gcc Addressspacs.c -o shel
hamsini@HamsiniVanam:~$ ./shell
PID: 696 Count: 1
PID: 697 Count: 1
PID: 696 Count: 2
PID: 697 Count: 2
PID: 696 Count: 3
PID: 697 Count: 3
PID: 696 Count: 4
PID: 697 Count: 4
PID: 697 Count: 5
PID: 696 Count: 5
hamsini@HamsiniVanam:~$

3.

hamsini@HamsiniVanam: ~

```
  GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

```
PID: 696 Count: 5
hamsini@HamsiniVanam:~$ nano paging.c
hamsini@HamsiniVanam:~$ gcc paging.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
hamsini@HamsiniVanam:~$
```

0

4.

```
  GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

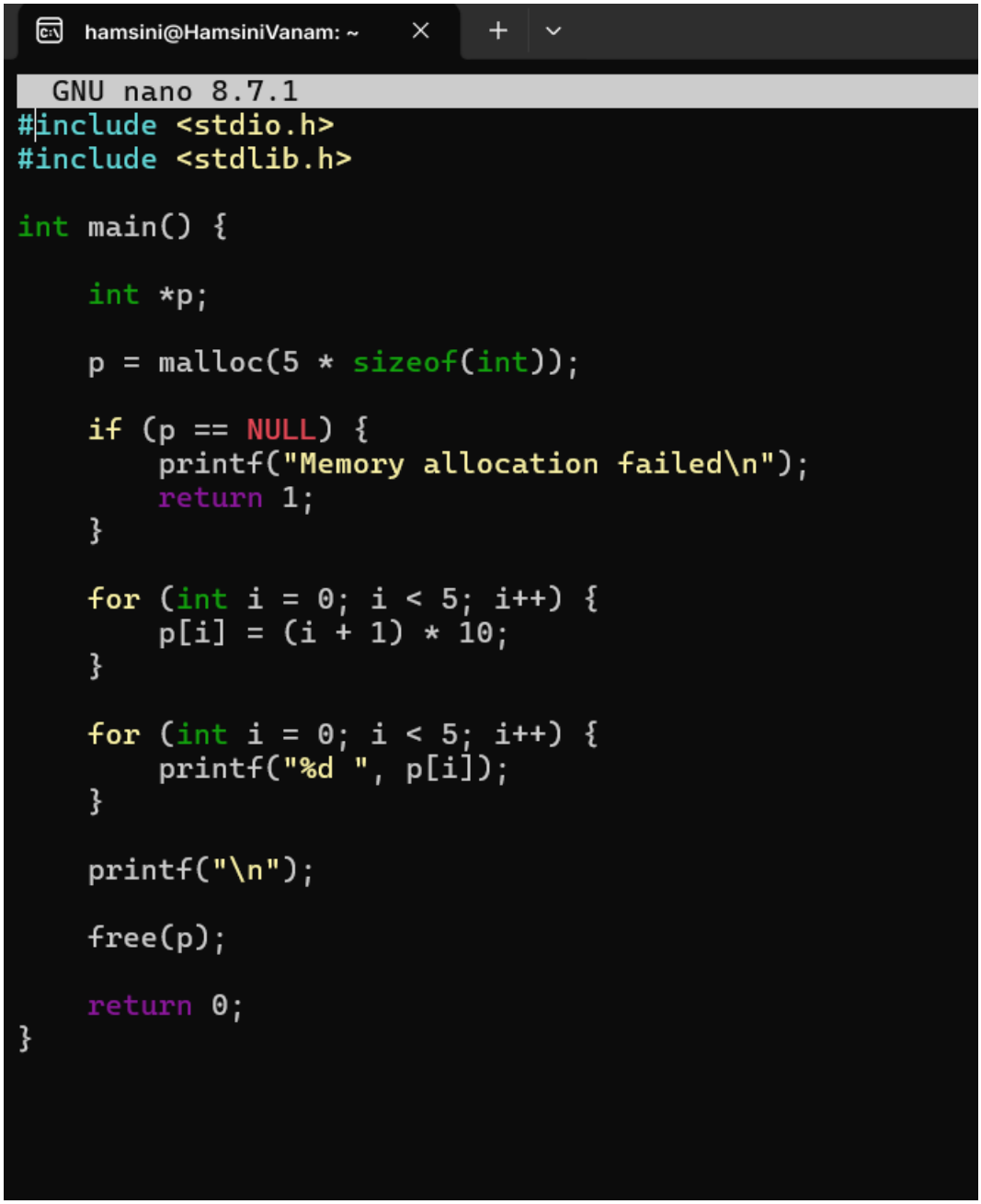
    int physical_address = frame * page_size + offset;

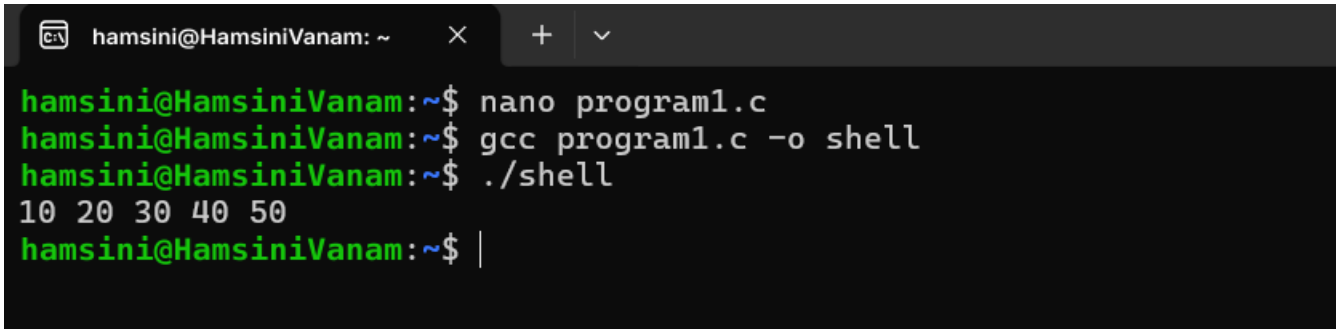
    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number     = %d\n", page);
    printf("Offset          = %d\n", offset);
    printf("Frame Number    = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}
```

```
hamsini@HamsiniVanam:~$ nano program.c
hamsini@HamsiniVanam:~$ gcc program.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Virtual Address = 250
Page Number     = 2
Offset          = 50
Frame Number    = 1
Physical Address = 150
hamsini@HamsiniVanam:~$
```

5.


hamsini@HamsiniVanam: ~
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
int main() {
int *p;
p = malloc(5 * sizeof(int));
if (p == NULL) {
printf("Memory allocation failed\n");
return 1;
}
for (int i = 0; i < 5; i++) {
p[i] = (i + 1) * 10;
}
for (int i = 0; i < 5; i++) {
printf("%d ", p[i]);
}
printf("\n");
free(p);
return 0;
}


hamsini@HamsiniVanam: ~
hamsini@HamsiniVanam:~$ nano program1.c
hamsini@HamsiniVanam:~$ gcc program1.c -o shell
hamsini@HamsiniVanam:~$ ./shell
10 20 30 40 50
hamsini@HamsiniVanam:~$

6.

hamsini@HamsiniVanam: ~

```
  GNU nano 8.7.1
#include <stdio.h>

int main() {

    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

```
hamsini@HamsiniVanam:~$ nano program2.c
hamsini@HamsiniVanam:~$ gcc program2.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Before accessing memory
Segmentation fault         ./shell
hamsini@HamsiniVanam:~$
```